

BAGHDAD

TAXI DEMAND PROJECT

TAXI DEMAND FORECASTING

Predicting hourly ride demand for eight Baghdad districts, one hour ahead — from raw trip logs to a live dashboard.

- **102,144 Zone-Hours**
- **18 Months**
Jan 2025 – Jun 2026
- **8 Districts**
Karkh & Rusafa



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STEP ONE: SQL

Cleaning and preparing data in sql for the next step in python

```
117
118 #=====
119
120 • CREATE TABLE trips_clean AS
121 SELECT
122     trip_id      ,
123     pickup_datetime ,
124     dropoff_datetime ,
125     pickup_zone   ,
126     dropoff_zone   ,
127     trip_status    ,
128     distance_km    ,
129     duration_min    ,
130     fare_iqd       ,
131     payment_type    ,
132     passenger_count ,
133     vehicle_type    ,
134     driver_id       ,
135     driver_rating
136 FROM (
137     SELECT r.trip_id , r.pickup_datetime , r.dropoff_datetime,
138            COALESCE(zm_p.clean_names, r.pickup_zone) AS pickup_zone,
139            COALESCE(zm_d.clean_names, r.dropoff_zone) AS dropoff_zone,
140            CASE WHEN LOWER(TRIM(r.trip_status)) LIKE 'complet%' THEN 'completed'
141                 WHEN LOWER(TRIM(r.trip_status)) LIKE '%driver%'
142                      AND LOWER(TRIM(r.trip_status)) LIKE '%cancel%' THEN 'cancelled_by_driver'
143                 WHEN LOWER(TRIM(r.trip_status)) LIKE '%rider%'
144                      AND LOWER(TRIM(r.trip_status)) LIKE '%cancel%' THEN 'cancelled_by_rider'
145                 WHEN LOWER(TRIM(r.trip_status)) LIKE '%no%driver%' THEN 'no_driver_found'
146                 ELSE LOWER(TRIM(r.trip_status))
147            END AS trip_status,
148            CASE WHEN r.distance_km BETWEEN 0.1 AND 100 THEN r.distance_km END AS distance_km,
149            CASE WHEN r.duration_min BETWEEN 1 AND 180 THEN r.duration_min END AS duration_min,
150            CASE WHEN r.fare_iqd > 0 THEN r.fare_iqd END AS fare_iqd, r.payment_type,
151            CASE WHEN r.passenger_count BETWEEN 1 AND 6 THEN r.passenger_count END AS passenger_count, r.vehicle_type, r.driver_id, r.driver_rating,
152            ROW_NUMBER() OVER(PARTITION BY r.trip_id ORDER BY r.pickup_datetime) AS rn
153     FROM raw_trips r
154     LEFT JOIN zone_name_map zm_p
155     ON zm_p.messy_names = LOWER(REGEXP_REPLACE(TRIM(r.pickup_zone), '[^A-Za-z]', ''))
156     LEFT JOIN zone_name_map zm d
```

Output

#	Time	Action	Message
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STEP TWO: PYTHON

EDA , Analyze and Training the model in python

```
Q Commands + Code + Text ▶ Run all
```

```
[ ]
    split_date = '2026-02-18'

    X_train = X[df['hour_ts'] < split_date]
    X_test = X[df['hour_ts'] >= split_date]

    y_train = y[df['hour_ts'] < split_date]
    y_test = y[df['hour_ts'] >= split_date]

    scaler = StandardScaler()
    X_train_scaled = scaler.fit_transform(X_train)
    X_test_scaled = scaler.transform(X_test)

    print("Train:" , X_train.shape , " Test:" , X_test.shape)
```

▼ Train: (76608, 36) Test: (25536, 36)

▼ Evaluate Baseline Model Performance

```
[ ]
    baseline_pred = X_test['lag_168']

    print('baseline performance:')
    print('-----')
    print('MAE:', mean_absolute_error(y_test ,baseline_pred))
    print('RMS:', np.sqrt(mean_squared_error(y_test ,baseline_pred)))
    print('R2:', r2_score(y_test ,baseline_pred))
```

▼ baseline performance:

MAE: 3.7313204887218046
RMS: 5.3757430855468264
R2: 0.47608470981990303

▼ Train and Evaluate XGBoost Model

```
[ ]
    xgb_model = XGBRegressor(random_state=42, n_estimators=500, learning_rate=0.05, max_depth=6)
    xgb_model.fit(X_train, y_train)

    xgb_pred = xgb_model.predict(X_test)
    xgb_pred = np.clip(xgb_pred, 0, None)

    print("XGBoost Model Performance:")
    print('-----')
    print("MAE:", mean_absolute_error(y_test, xgb_pred))
    print("RMSE:", np.sqrt(mean_squared_error(y_test, xgb_pred)))
    print("R2:", r2_score(y_test, xgb_pred))
```

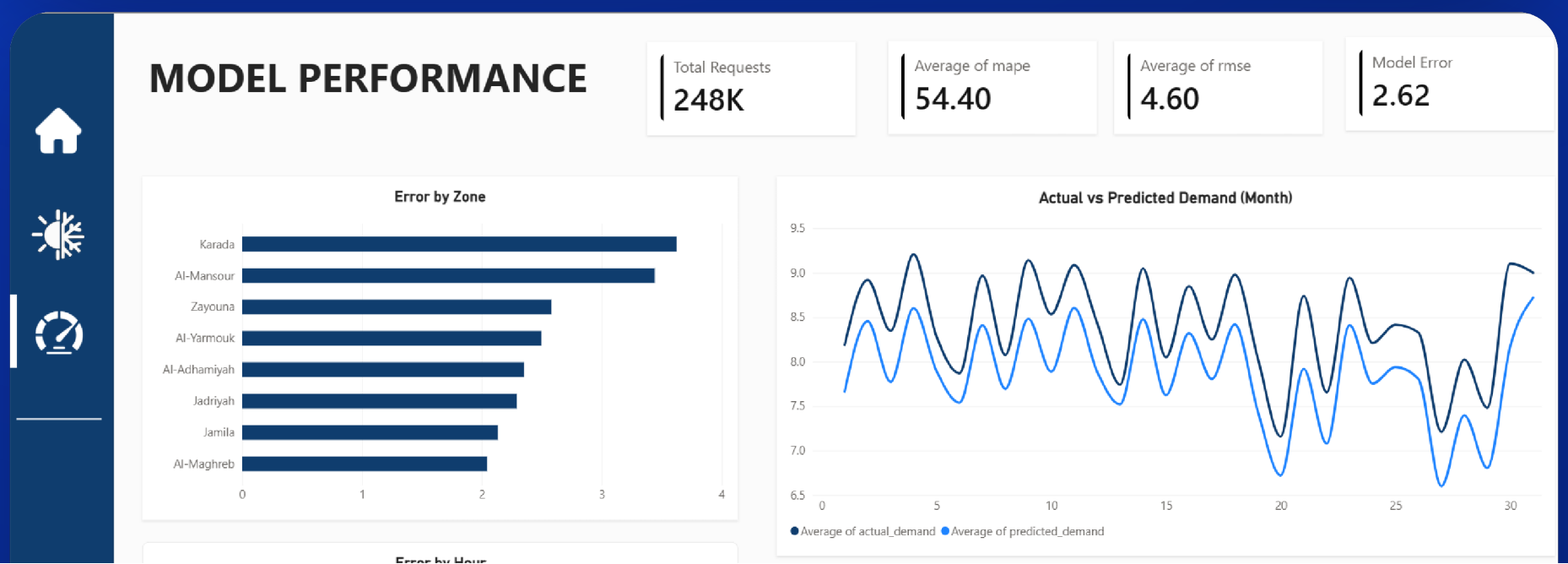
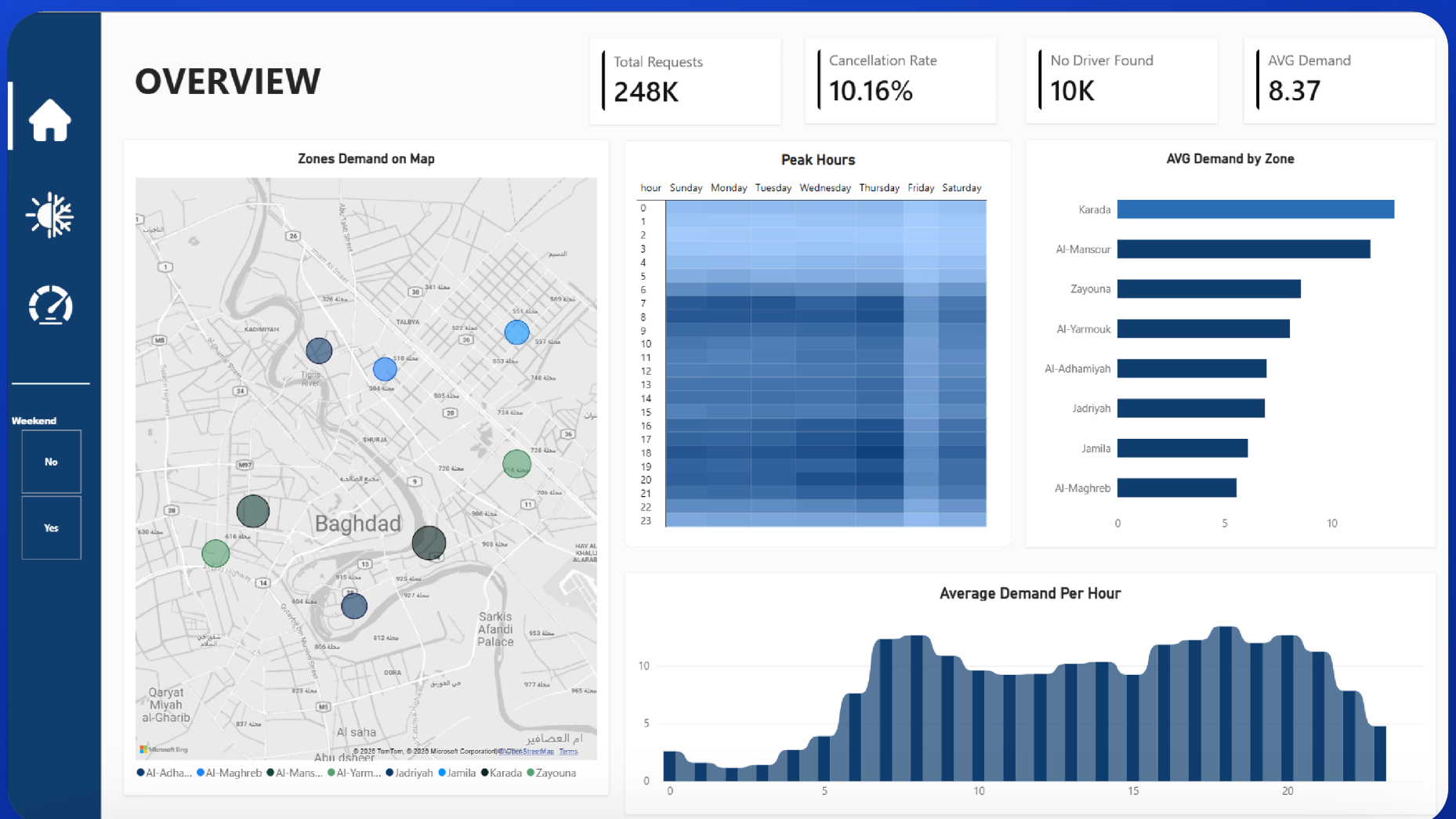
▼ XGBoost Model Performance:

MAE: 2.620990037918091
RMSE: 3.81043625303935
R2: 0.7367711663246155

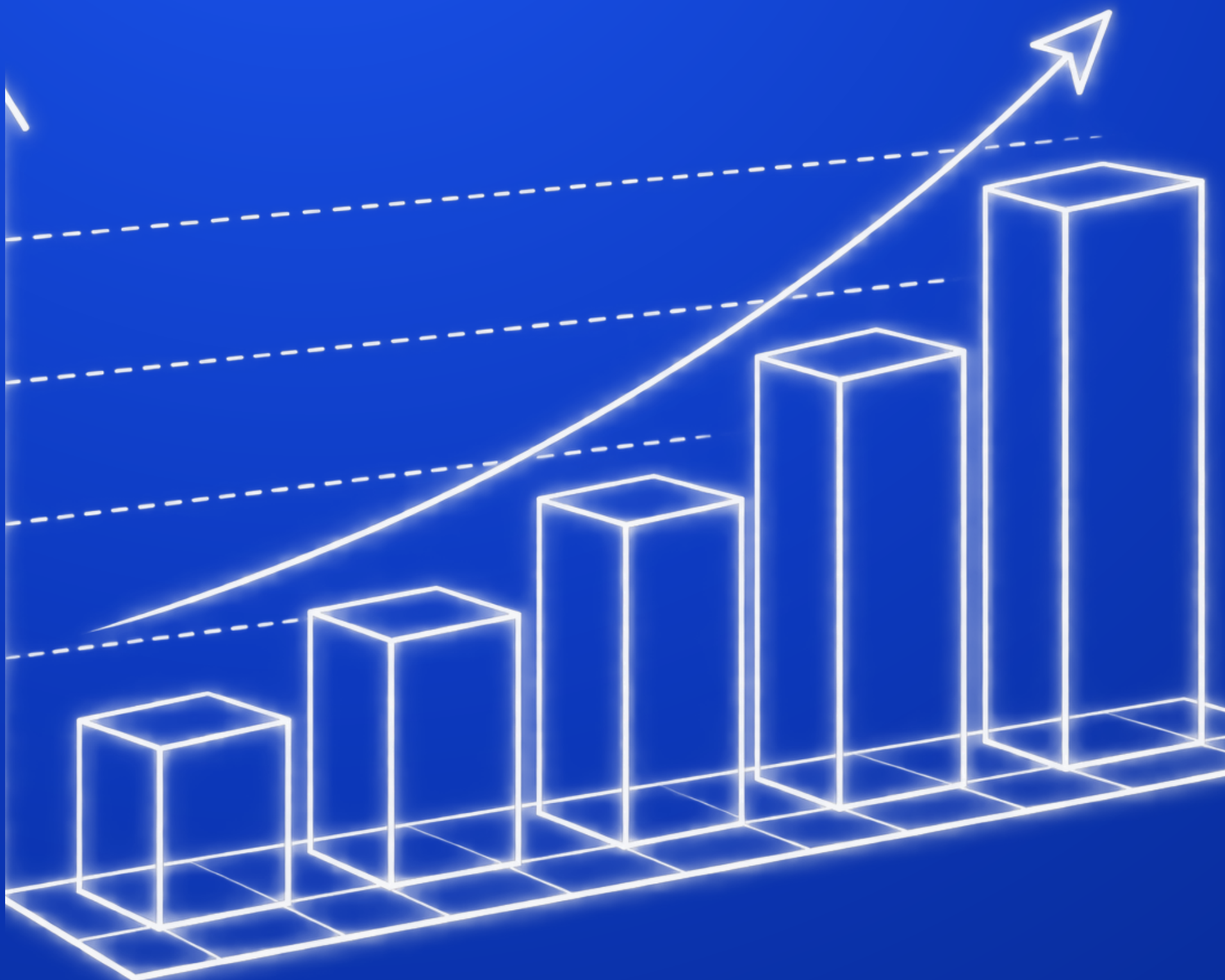


STEP THREE POWER BI

Building a interactive dashboard for Overview Demand, Conditions and Model Performance

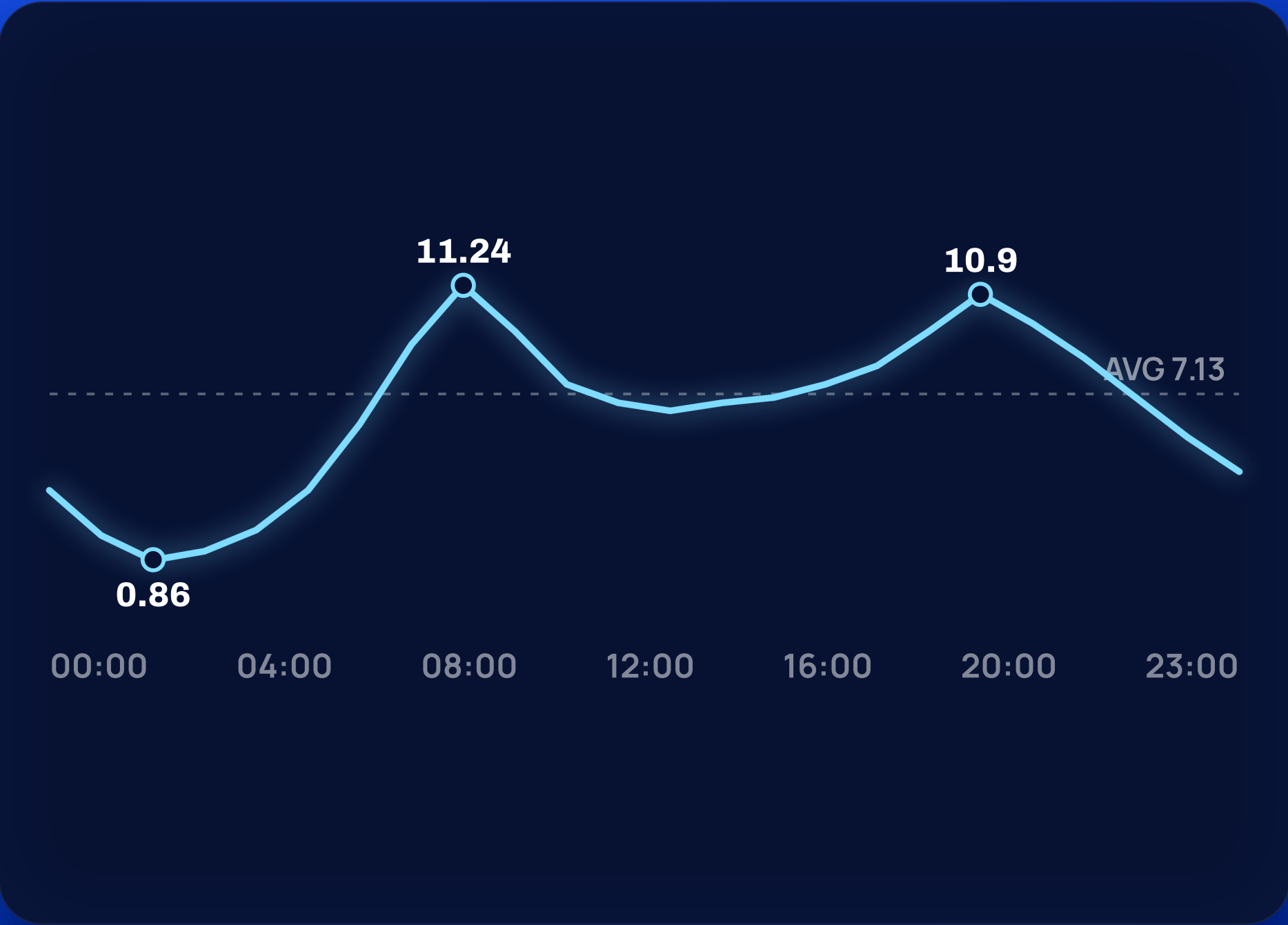


INSIGHTS AND FINDINGS



DEMAND BY HOUR OF DAY

Averaged across eight districts and 4.5 months.



11.24
MORNING PEAK
08:00, the commute

0.86
OVERNIGHT TROUGH
02:00, 13× quieter

DEMAND BY DISTRICT



MODEL VS. BASELINE

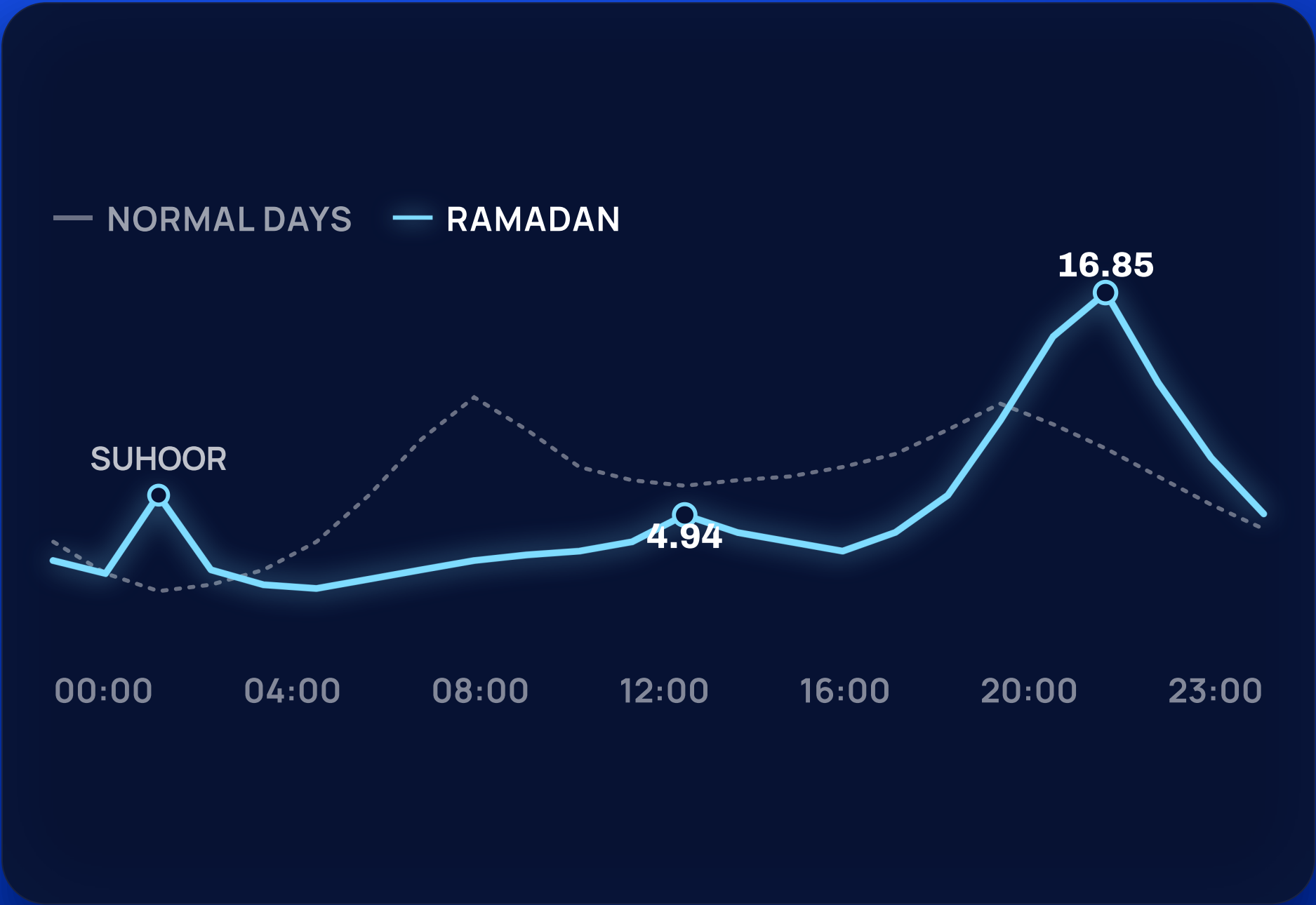
Scored on unseen data from 18 Feb 2026 onward — a window that contains Ramadan, both Eids and Ashura.

	BASELINE SAME HOUR LAST WEEK	XGBOOST
MAE	3.727	2.630
RMSE	5.370	3.827
R ²	0.476	0.734
MAPE	65.3%	43.5%

29.4% lower mean absolute error than repeating last week's demand.

RAMADAN CHANGES THE PATTERN

Daytime demand collapses, an iftar rush builds after sunset, and a suhoor spike appears near 02:00.



16.85
PEAK, 20:00

4.94
MIDDAY, 12:00

2.83/2.57
MAE,
RAMADAN/NORMAL

CALENDAR & WEATHER

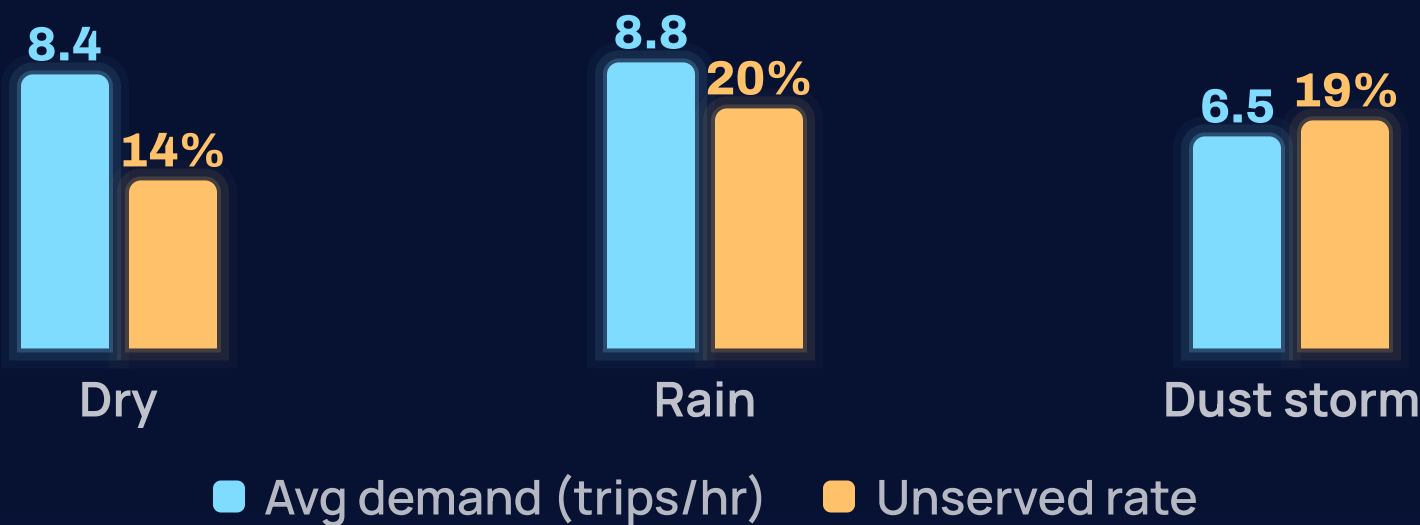
Every calendar event, compared

Ramadan and Eid barely move the daily average – their effect is about when demand happens, not how much.



Weather's real cost isn't demand

Rain barely lifts ride requests, but the unserved rate jumps.



THANK YOU

Hassan M. Ahmed